Features

- $(VIX/100)^2$ is a popular estimator the future 30-day realized variance of SPX
- By CBOE's definition of VIX, when there are 30-day options trading, we have

$$(VIX/100)^2 = \sum_h 2\Delta K_h e^{R_{30}T_{30}} \frac{O(K_h, T_{30})}{K_h^2}$$

- Use normalized $O(K_h, T_i)/K_h^2$ as features X
- Variance Risk Premium

$$VRP = VIX_t^{*2} - Var_t$$

- How many strikes to consider?
  - o VIX: liquid and illiquid ones
  - o VIX*: $2n+1$

Data and Feature Engineering

- Bloomberg sourced Implied Volatility (IV)
  - 10, 25, 40 delta and ATM (7 strikes)
  - 30D maturity
- 1M Libor, SPX spot (and historical vol), VIX spot
- From Jan 2005 to Feb 2021
- Calculate implied strikes for Black-Scholes, e.g., for 10 delta, solve *K* in

```
@BQL("SPX Index","dropna(implied_volatility(expiry=30D,delta=25,
Call_Put = Call ))","dates=range(-20y,0d)","cols=2;rows=4070")
```

$$N\left(\frac{\log\left(S_0/K\right)+\left(R_{30}+\sigma_{30}^2/2\right)T_{30}}{\sigma_{30}\sqrt{T_{30}}}\right)=10\%.$$